H2RES

Highway to Renewable Energy Systems

H2RES v1.1 Report (Draft)

Mathematical Formulation and Data Dictionary

Document Type: Technical Document

Reviewed by: ......

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1 Scope

This document reconstructs the optimization model implemented in `scripts/Build_model.py` and the data processing logic in `scripts/read_process_data.py`. It lists all decision variables and all symbols used as parameters/sets/auxiliary terms, then explains the objective and each constraint block.

2 Model Architecture Diagram

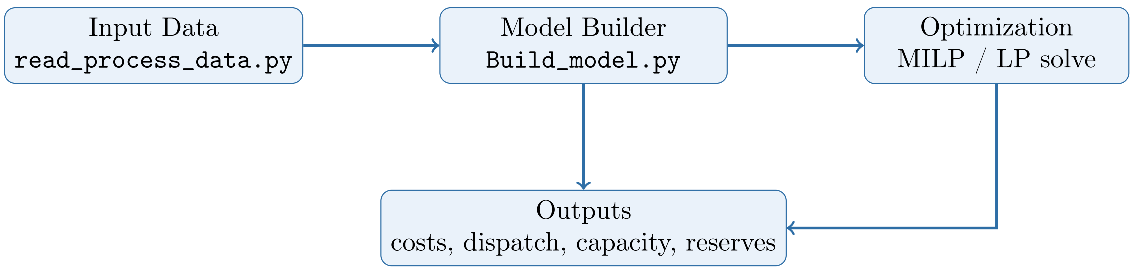

Figure 1: High-level workflow of the H2RES optimization model.

3 General Changes Introduced in v1.1

This summary highlights the main formulation-level differences introduced in the current v1.1 implementation, while the detailed blue additions throughout the rest of the document preserve equation-level traceability.

- The v1.1 model introduces explicit unmet-service variables for electricity, district heating, and general/industrial heat, instead of relying only on the legacy `ceep` slack representation.
- The objective function now includes explicit scarcity and unmet-service penalty terms, so adequacy relaxation is monetized more transparently in the optimization problem.
- Two new scenario controls affect system behavior directly: `ceep_penalty` activates monetary valuation of CEEP, and `fossil_dispatch` can disable dispatch from the coal/gas/diesel subset.
- Thermal and electricity balance closures are extended accordingly, so unmet demand can appear explicitly in the implemented balance equations rather than remaining implicit in a single aggregate slack construct.
- The preprocessing layer now reads the active v1.1 data package from CSV/XLSX sources and builds technology-specific annual CAPEX trajectories for major flexibility technologies, while preserving the same identifier-matching logic across years, periods, units, and fuels.
- The current v1.1 code also contains a reduced time-horizon definition (`periods = list(range(1,10))`), which should be interpreted as an implementation-specific setting unless it is intentionally part of the study design.

4 Data Dictionary Tables (Complete)

4.1 Decision Variables (Complete)

Variable (original model notation)	Type	Unit	Description
$\texttt{rampup\_cost}_{disp\_units,t,y}$	Cost auxiliary	EUR	Auxiliary upward ramping cost variable for dispatchable units.
$\texttt{rampdown\_cost}_{disp\_units,t,y}$	Cost auxiliary	EUR	Auxiliary downward ramping cost variable for dispatchable units.
$\texttt{generation}_{u,t,y}$	Operation	MW_e (hourly average)	Hourly electric generation by unit.
$\texttt{imports}_{t,y}$	Operation	MW_e (hourly average)	Hourly electricity imports.
$\texttt{ceep}_{t,y}$	Operation	MW_e (hourly average)	Curtailment / non-served energy slack variable (ceep).
$\texttt{noload\_elec}_{t,y}$	Operation	MW_e (hourly average)	Explicit unmet-electricity-service variable introduced in the v1.1 implementation.
$\texttt{cap\_inv}_{u,y}$	Investment	MW_e	Annual generation capacity investment.
$\texttt{CO2\_total}_{y}$	Emissions	tCO2/year	Annual CO2 emissions from electric generation.
$\texttt{CO2\_total\_heat}_{y}$	Emissions	tCO2/year	Annual CO2 emissions from heat supply.
$\texttt{CO2\_total\_industry}_{y}$	Emissions	tCO2/year	Annual CO2 emissions from industrial fuels.
$\texttt{cap\_inv\_FC}_{FC\_units,y}$	Investment	MW_e	Annual fuel-cell capacity investment.
$\texttt{generation\_FC}_{FC\_units,t,y}$	Operation	MW_e (hourly average)	Hourly electric generation from fuel cells.
$\texttt{storage\_level}_{hydro\_storage\_units,t,y}$	State variable	MWh_e	Hydro storage level (HDAM/HPHS).
$\texttt{sto\_out\_flow}_{hydro\_storage\_units,t,y}$	Operation	MW_e eq.	Hydro spill / outflow not converted to electricity.
$\texttt{heat\_gen}_{chp\_units,t,y}$	Operation	MW_th	Heat output from CHP units.
$\texttt{noload\_heat\_dh}_{chp\_units,t,y}$	Operation	MW_th	Explicit unmet district-heating service variable used in the v1.1 demand closure.
$\texttt{noload\_heat\_ind}_{t,y}$	Operation	MW_th	Explicit unmet general/industrial heat-service variable used in the v1.1 demand closure.
$\texttt{heat\_input}_{chp\_units,t,y}$	Operation	MW_th eq.	Electric-to-heat extraction input in CHP coupling.
$\texttt{heat\_sto\_level}_{chp\_units,t,y}$	Operation	MWh_th	CHP thermal storage state of charge.
$\texttt{Fuel\_ind}_{industry\_fuels,t,y}$	Operation	MW_th	Industrial fuel use by fuel type.
$\texttt{H2tInd}_{t,y}$	Operation	MW_H2	Hydrogen use in industrial high-temperature demand.
$\texttt{heat\_pump\_out}_{HeatPump\_units,t,y}$	Operation	MW_th	Heat-pump operation or expansion variable by market/segment.
$\texttt{heat\_pump\_dh}_{HeatPump\_units,chp\_units,t,y}$	Operation	MW_th	Heat-pump operation or expansion variable by market/segment.
$\texttt{heat\_pump\_ind}_{HeatPump\_units,t,y}$	Operation	MW_th	Heat-pump operation or expansion variable by market/segment.
$\texttt{heat\_pump\_dh\_cap}_{HeatPump\_units,chp\_units,y}$	Investment	MW_th	Heat-pump operation or expansion variable by market/segment.
$\texttt{heat\_pump\_ind\_cap}_{HeatPump\_units,y}$	Investment	MW_th	Heat-pump operation or expansion variable by market/segment.
$\texttt{cooling\_pump\_dh}_{HeatPump\_units,chp\_units,t,y}$	Operation	MW_cool	Cooling output from heat-pump technologies.
$\texttt{cooling\_pump\_ind}_{HeatPump\_units,t,y}$	Operation	MW_cool	Cooling output from heat-pump technologies.
$\texttt{boiler\_dh\_generation}_{thermal\_units,chp\_units,t,y}$	Operation	MW_th	Thermal boiler heat production variable.
$\texttt{boiler\_ind\_generation}_{thermal\_units,t,y}$	Operation	MW_th	Thermal boiler heat production variable.
$\texttt{trad\_boiler\_ind\_cap\_inv}_{thermal\_units,y}$	Investment	MW_th	Thermal boiler capacity investment variable.
$\texttt{trad\_boiler\_dh\_cap\_inv}_{thermal\_units,chp\_units,y}$	Investment	MW_th	Thermal boiler capacity investment variable.

Variable (original model notation)	Type	Unit	Description
`ResToHeat`$_{t,y}$	Operation	MW_e (hourly average)	Electricity converted to heat/cooling services (power-to-heat).
`heat_pump_cap`$_{HeatPump\_units,y}$	Investment	MW_th	Heat-pump operation or expansion variable by market/segment.
`heat_pump_sto_soc`$_{HeatPump\_units,t,y}$	State variable	MWh_th	Heat-pump operation or expansion variable by market/segment.
`ev_sto_in`$_{t,y}$	Operation	MW_e	EV fleet charging/discharging/storage operation variable.
`ev_sto_out`$_{t,y}$	Operation	MW_e	EV fleet charging/discharging/storage operation variable.
`ev_sto_soc`$_{t,y}$	State variable	MWh_e	Aggregate EV fleet state of charge.
`stat_sto_in`$_{Stat\_storage\_unit,t,y}$	Operation	MW_e	Stationary battery operation/investment variable.
`stat_sto_out`$_{Stat\_storage\_unit,t,y}$	Operation	MW_e	Stationary battery operation/investment variable.
`stat_sto_soc`$_{Stat\_storage\_unit,t,y}$	State variable	MWh_e	Stationary battery state of charge.
`stat_sto_cap_inv`$_{Stat\_storage\_unit,y}$	Investment	MWh_e	Stationary battery operation/investment variable.
`power_to_h2`$_{t,y}$	Operation	MW_e	Total electric load used for hydrogen production.
`h2_sto_soc`$_{H2\_storage\_unit,t,y}$	State variable	MWh_H2	Hydrogen storage state of charge.
`h2_sto_out`$_{H2\_storage\_unit,t,y}$	Operation	MW_H2	Hydrogen storage operation/investment variable.
`h2_sto_max_cap`$_{H2\_storage\_unit,y}$	Investment	MWh_H2	Hydrogen storage operation/investment variable.
`h2_electrolysis_max_cap`$_{Elec\_h2\_units,y}$	Investment	MW_H2	Decision variable used in the optimization model.
`generation_H2`$_{Elec\_h2\_units,t,y}$	Operation	MW_H2	Hydrogen production by electrolyzers.
`tot_electric_load`$_{t,y}$	Auxiliary	MW_e	Decision variable used in the optimization model.
`max_cap_installed`$_{y}$	Auxiliary	MW_e	Decision variable used in the optimization model.

4.2 Parameters, Sets, and Auxiliary Terms (Complete)

Parameter / set (original model notation)	Category	Unit	Description
`Boiler_decom_rate`$_{thermal\_units}$	Technical	p.u. per hour/year	Capacity decommissioning/degradation rate.
`Boiler_decom_start_new`$_{thermal\_units}$	Technical	year	Thermal boiler module parameter.
`CHPMaxHeat`$_{chp\_units}$	Technical	MW	CHP coupling / thermal storage parameter.
`CHPPowerLossFactor`$_{chp\_units}$	Technical	p.u.	CHP coupling / thermal storage parameter.
`CHPPowerToHeat`$_{chp\_units}$	Technical	p.u.	CHP coupling / thermal storage parameter.
`CO2_cost_industry`	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
`CO2_emissions_cost`	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
`ceep_cost`	Objective auxiliary	EUR	Conditional scarcity-cost term activated when `ceep_penalty=True`.
`no_load_cost`	Objective auxiliary	EUR	Penalty term for the explicit electricity and general-heat unmet-service variables in v1.1.
`no_load_cost_dh`	Objective auxiliary	EUR	Penalty term for explicit unmet district-heating service in v1.1.
`CO2_factor`$_{u}$	Technical	tCO2/MWh	CO2 intensity factor.
`CO2_limit`$_{y}$	Policy	tCO2/year	Policy parameter by model year.

Parameter / set (original model notation)	Category	Unit	Description
`CO2_price`$_{y}$	Policy	EUR/tCO2	Policy parameter by model year.
`Elec_h2_units`	Set	-	Index set/subset used in constraints and objective terms.
`FC_eff`$_{FC\_units}$	Technical	p.u.	Hydrogen and fuel-cell module parameter.
`FC_init_cap`$_{FC\_units}$	Technical	MW	Hydrogen and fuel-cell module parameter.
`FC_max_cap`$_{FC\_units}$	Technical	MW	Hydrogen and fuel-cell module parameter.
`FC_units`	Set	-	Index set/subset used in constraints and objective terms.
`FC_var_cost`$_{FC\_units}$	Technical	EUR/MWh	Hydrogen and fuel-cell module parameter.
`Fossil_hydro`	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
`Fuel_CO2`$_{industry\_fuels}$	Technical	tCO2/MWh	CO2 intensity factor.
`H2_demand`$_{demand\_sectors,t,y}$	Demand	MW_H2	Exogenous demand trajectory by market/sector and time.
`H2_eff`$_{Elec\_h2\_units}$	Technical	p.u.	Hydrogen and fuel-cell module parameter.
`H2_init_cap`$_{Elec\_h2\_units}$	Technical	MW	Hydrogen and fuel-cell module parameter.
`H2_max_cap`$_{Elec\_h2\_units}$	Technical	MW	Hydrogen and fuel-cell module parameter.
`H2_sto_init`$_{H2\_storage\_unit}$	Technical	MWh	Hydrogen and fuel-cell module parameter.
`H2_sto_max_cap`$_{H2\_storage\_unit}$	Technical	MWh	Hydrogen and fuel-cell module parameter.
`H2_storage_unit`	Technical	model units	Hydrogen and fuel-cell module parameter.
`H2_var_cost`$_{Elec\_h2\_units}$	Technical	EUR/MWh	Hydrogen and fuel-cell module parameter.
`HP_Boiler_dh_init_cap`$_{HeatPump\_units,chp\_units}$	Technical	MW	Heat pump / heat system parameter.
`HP_COP_period`$_{HeatPump\_units,t,y}$	Technical	p.u.	Heat pump / heat system parameter.
`HP_Decom_ind`$_{HeatPump\_units,y}$	Technical	p.u.	Heat pump / heat system parameter.
`HP_decom_rate`$_{HeatPump\_units}$	Technical	p.u. per hour/year	Capacity decommissioning/degradation rate.
`HP_decom_start_new`$_{HeatPump\_units}$	Technical	year	Heat pump / heat system parameter.
`HP_ind_init_cap`$_{HeatPump\_units}$	Technical	MW	Heat pump / heat system parameter.
`HP_max_cap`$_{HeatPump\_units}$	Technical	MW	Heat pump / heat system parameter.
`HP_sto_init`$_{HeatPump\_units}$	Technical	MWh	Heat pump / heat system parameter.
`HP_var_cost`$_{HeatPump\_units}$	Technical	model units	Heat pump / heat system parameter.
`HeatPump_units`	Set	-	Index set/subset used in constraints and objective terms.
`Imp_price_inc`$_{y}$	Economic	% per year	Exogenous economic parameter.
`Import_Price`$_{t,y}$	Economic	EUR/MWh_e	Exogenous economic parameter.
`Industry_HT_demand`$_{t,y}$	Demand	MW-equivalent (hourly)	Exogenous demand trajectory by market/sector and time.
`NPV`	Economic	p.u.	Exogenous economic parameter.
`NoResToHeatInv`	Scenario flag	bool	Boolean switch that activates/deactivates a model block.
`ResToHeat_cost`	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
`STOCapacity_heat`$_{chp\_units}$	Technical	MWh	CHP coupling / thermal storage parameter.
`StaSto_Decom_ind`$_{y}$	Technical	p.u.	Exogenous parameter or helper alias used in constraints/objective.
`Sta_sto_decom_rate`$_{Stat\_storage\_unit}$	Technical	p.u. per hour/year	Capacity decommissioning/degradation rate.
`Sta_sto_decom_start_new`$_{Stat\_storage\_unit}$	Technical	year	Stationary storage module parameter.
`Sta_sto_eff`$_{Stat\_storage\_unit}$	Technical	p.u.	Stationary storage module parameter.
`Sta_sto_init_cap`$_{Stat\_storage\_unit}$	Technical	MWh	Stationary storage module parameter.
`Sta_sto_max_cap`$_{Stat\_storage\_unit}$	Technical	MWh	Stationary storage module parameter.
`Sta_sto_var_cost`$_{Stat\_storage\_unit}$	Technical	model units	Stationary storage module parameter.

Parameter / set (original model notation)	Category	Unit	Description
`Stat_storage_unit`	Technical	model units	Exogenous parameter or helper alias used in constraints/objective.
`TechChange`	Technical	p.u.	Capacity expansion/decommissioning trajectory parameter.
`Thermal_CO2`$_{thermal\_units}$	Technical	tCO2/MWh	CO2 intensity factor.
`Thermal_Decom_DH`$_{thermal\_units,chp\_units,y}$	Technical	p.u.	Thermal boiler module parameter.
`Thermal_Decom_ind`$_{thermal\_units,y}$	Technical	p.u.	Thermal boiler module parameter.
`Thermal_ind_init_cap`$_{thermal\_units}$	Technical	MW	Thermal boiler module parameter.
`Thermal_max_cap`$_{thermal\_units}$	Technical	MW	Thermal boiler module parameter.
`V2G_cost`$_{t,y}$	Economic	EUR/MWh_e	Exogenous economic parameter.
`aval_factor_plant`$_{u,t,y}$	Technical	p.u.	Hydro inflow/efficiency/availability parameter.
`biomass_units`	Set	-	Index set/subset used in constraints and objective terms.
`boiler_gen_cost`	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
`cap_factor`$_{u}$	Technical	p.u.	Technical/economic parameter for generation technologies.
`cap_inv_cost`$_{u,y}$	Technical	EUR/MW	Technical/economic parameter for generation technologies.
`carbonLimit`	Scenario flag	bool	Boolean switch that activates/deactivates a model block.
`ceep_demand`$_{y}$	Technical	model units	Exogenous parameter or helper alias used in constraints/objective.
`ceep_limit`	Scenario flag	bool	Boolean switch that activates/deactivates a model block.
`ceep_penalty`	Scenario flag	bool	Boolean switch that activates/deactivates monetization of CEEP in the objective.
`ceep_value`	Policy	EUR/MWh_e	Scarcity-value parameter applied to CEEP when the penalty switch is active.
`ceep_parameter`$_{y}$	Policy	p.u.	Upper bound on annual ceep as share of annual demand.
`chp_units`	Set	-	Index set/subset used in constraints and objective terms.
`cooling_demand`$_{demand\_sectors,t,y}$	Demand	MW-equivalent (hourly)	Exogenous demand trajectory by market/sector and time.
`decom_rate`$_{u}$	Technical	p.u. per hour/year	Capacity decommissioning/degradation rate.
`decom_start_new`$_{u}$	Technical	year	Capacity expansion/decommissioning trajectory parameter.
`decom_start_old`$_{u}$	Technical	year	Capacity expansion/decommissioning trajectory parameter.
`demand`$_{demand\_sectors,t,y}$	Demand	MW-equivalent (hourly)	Exogenous demand trajectory by market/sector and time.
`demand_sectors`	Technical	model units	Exogenous parameter or helper alias used in constraints/objective.
`disp_units`	Set	-	Index set/subset used in constraints and objective terms.
`evToPowert_cost`	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
`ev_Demand`$_{y}$	Technical	MW-equivalent (hourly)	EV/V2G module parameter.
`ev_Grid_P_max`$_{y}$	Technical	MW	EV/V2G module parameter.
`ev_Grid_eff`$_{y}$	Technical	p.u.	EV/V2G module parameter.
`ev_availability`$_{t,y}$	Technical	p.u.	EV/V2G module parameter.
`ev_sto_max`$_{y}$	Technical	MWh	EV/V2G module parameter.
`ev_sto_min`$_{y}$	Technical	p.u.	EV/V2G module parameter.
`ev_transp_load`$_{t,y}$	Technical	MW-equivalent (hourly)	EV/V2G module parameter.
`exports`$_{t,y}$	Technical	MW_e	Interconnection/import-export parameter by period.
`fossil_dispatch`	Scenario flag	bool	Boolean switch that can force the coal/gas/diesel subset out of dispatch.
`LostLoad_Cost`	Economic	EUR/MWh	Penalty coefficient used for explicit unmet-service variables in the v1.1 implementation.

Parameter / set (original model notation)	Category	Unit	Description
`fossil_units`	Set	-	Index set/subset used in constraints and objective terms.
`coal_gas_diesel_units`	Set	-	Subset of fossil units affected by the optional `fossil_dispatch` shutdown logic in v1.1.
`fuel_share_max`$_{t,y,industry\_fuels}$	Technical	p.u.	Exogenous parameter or helper alias used in constraints/objective.
`h2_CAPEX_cost`	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
`h2_OPEX_cost`	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
`h2_storage_cost`	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
`hdam_efficiency`$_u$	Technical	p.u.	Hydro inflow/efficiency/availability parameter.
`hdam_inflow`$_{u,t,y}$	Technical	inflow p.u.	Hydro inflow/efficiency/availability parameter.
`hdam_max_storage`$_u$	Technical	MWh	Hydro inflow/efficiency/availability parameter.
`hdam_units`	Set	-	Index set/subset used in constraints and objective terms.
`heat_demand`$_{chp\_units,t,y}$	Demand	MW-equivalent (hourly)	Exogenous demand trajectory by market/sector and time.
`heat_pump_cap_cost`	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
`heat_var_cost`$_{HeatPump\_units}$	Technical	EUR/MWh	Heat pump / heat system parameter.
`hphs_efficiency`$_u$	Technical	p.u.	Hydro inflow/efficiency/availability parameter.
`hphs_inflow`$_{u,t,y}$	Technical	inflow p.u.	Hydro inflow/efficiency/availability parameter.
`hphs_max_storage`$_u$	Technical	MWh	Hydro inflow/efficiency/availability parameter.
`hphs_units`	Set	-	Index set/subset used in constraints and objective terms.
`hror_units`	Set	-	Index set/subset used in constraints and objective terms.
`hydro_storage`	Scenario flag	bool	Boolean switch that activates/deactivates a model block.
`hydro_storage_units`	Set	-	Index set/subset used in constraints and objective terms.
`hydro_units`	Set	-	Index set/subset used in constraints and objective terms.
`import_cost`	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
`import_ntc`$_t$	Technical	MW_e	Interconnection/import-export parameter by period.
`industry_fuel_cost`	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
`industry_fuels`	Technical	model units	Exogenous parameter or helper alias used in constraints/objective.
`industry_fuels_price`$_{industry\_fuels,t,y}$	Technical	EUR/MWh_fuel	Exogenous parameter or helper alias used in constraints/objective.
`inv_cost`	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
`inv_cost_FuelCell`	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
`inv_cost_boiler`	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
`inv_cost_fossil`	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
`max_inv_period`$_u$	Technical	MW	Capacity expansion/decommissioning trajectory parameter.
`max_load`$_u$	Technical	MW_e	Technical/economic parameter for generation technologies.
`nondisp_units`	Set	-	Index set/subset used in constraints and objective terms.
`per_mw`	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.

Parameter / set (original model notation)	Category	Unit	Description
per_mw_FC	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
periods	Set	-	Index set/subset used in constraints and objective terms.
ramp_cost$_u$	Technical	EUR/MW	Ramping capability/cost parameter.
ramp_down_cost_var	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
ramp_down_rate$_u$	Technical	p.u. per hour/year	Ramping capability/cost parameter.
ramp_up_cost_var	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
ramp_up_rate$_u$	Technical	p.u. per hour/year	Ramping capability/cost parameter.
rps_inv	Scenario flag	bool	Boolean switch that activates/deactivates a model block.
statCapInvCost	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
statToPowert_cost	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
Elec_cap_costs_years$_{Elec_h2_units,y}$	Technical	EUR/MW	Annual electrolyzer CAPEX trajectory read from the v1.1 technology-cost workbook.
FC_cap_costs_years$_{FC_units,y}$	Technical	EUR/MW	Annual fuel-cell CAPEX trajectory read from the v1.1 technology-cost workbook.
HP_cap_costs_years$_{HeatPump_units,y}$	Technical	EUR/MW	Annual heat-pump CAPEX trajectory read from the v1.1 technology-cost workbook.
boiler_cap_costs_years$_{thermal_units,y}$	Technical	EUR/MW	Annual thermal-boiler CAPEX trajectory read from the v1.1 technology-cost workbook.
H2sto_costs_years$_{H2_storage_unit,y}$	Technical	EUR/MWh	Annual hydrogen-storage CAPEX trajectory read from the v1.1 technology-cost workbook.
battery_costs_years$_{Stat_storage_unit,y}$	Technical	EUR/MWh	Annual stationary-battery CAPEX trajectory read from the v1.1 technology-cost workbook.
thermal_units	Set	-	Index set/subset used in constraints and objective terms.
units	Technical	model units	Exogenous parameter or helper alias used in constraints/objective.
var_cost$_u$	Technical	EUR/MWh	Technical/economic parameter for generation technologies.
years	Set	-	Index set/subset used in constraints and objective terms.

Unit note: The model is hourly. Several power and energy quantities are numerically linked within each hour (MW average vs MWh per hour). Units above follow the most interpretable convention for each symbol.

5 Formulation by Equation Blocks and Their Role

Compact notation: $u \in \texttt{units}$, $t \in \texttt{periods}$, $y \in \texttt{years}$, $G_{u,t,y} = \texttt{generation}_{u,t,y}$, $I_{t,y} = \texttt{imports}_{t,y}$, $C_{t,y} = \texttt{ceep}_{t,y}$.

5.1 1) Objective Function

Description: This block aggregates all discounted annual cost components (operation, investment, flexibility, hydrogen, heat, and policy-related terms) into a single scalar objective Z.

$$\begin{aligned}\min\ Z = {} & \texttt{per\_mw} + \texttt{import\_cost} + \texttt{inv\_cost} + \texttt{inv\_cost\_fossil} \\ & + \texttt{ramp\_up\_cost\_var} + \texttt{ramp\_down\_cost\_var} + \texttt{heat\_pump\_cap\_cost} \\ & + \texttt{ResToHeat\_cost} + \texttt{boiler\_gen\_cost} + \texttt{inv\_cost\_boiler} + \texttt{CO2\_emissions\_cost} \\ & + \texttt{evToPowert\_cost} + \texttt{statToPowert\_cost} + \texttt{statCapInvCost} \\ & + \texttt{h2\_storage\_cost} + \texttt{h2\_CAPEX\_cost} + \texttt{h2\_OPEX\_cost} \\ & + \texttt{ceep\_cost} + \texttt{no\_load\_cost} + \texttt{no\_load\_cost\_dh} \\ & + \texttt{inv\_cost\_FuelCell} + \texttt{per\_mw\_FC} + \texttt{industry\_fuel\_cost}\end{aligned} \tag{1}$$

Role: minimizes total discounted system cost across electricity, heat/cooling, hydrogen, transport flexibility, and investment decisions. In the current v1.1 code, this role is extended by explicit scarcity penalties for CEEP and for unmet electricity/heat service.

5.2 2) Policy and Scenario Activation Blocks

Description: These logical switches activate or deactivate optional model modules through equivalent algebraic constraints, enabling scenario analysis without changing the model core.

$$\texttt{NoResToHeatInv=True} \Rightarrow \texttt{heat\_pump\_cap} = 0 \tag{2}$$

$$\texttt{NoResToHeatInv=True} \Rightarrow \texttt{ResToHeat} = 0 \tag{3}$$

$$\texttt{NoResToHeatInv=True} \Rightarrow \texttt{heat\_pump\_out} = 0 \tag{4}$$

$$\texttt{rps\_inv=True} \Rightarrow \text{RPS constraint is active} \tag{5}$$

$$\texttt{carbonLimit=True} \Rightarrow \text{annual CO2 cap is active} \tag{6}$$

$$\texttt{PrimaryReserve/SecondaryReserve=True} \Rightarrow \text{reserve modules are active} \tag{7}$$

$$\texttt{ceep\_penalty=True} \Rightarrow \text{CEEP is monetized in the objective} \tag{8}$$

$$\texttt{fossil\_dispatch=False} \Rightarrow \texttt{generation}_{u,t,y} = 0,\ u \in \texttt{coal\_gas\_diesel\_units} \tag{9}$$

Role: allows policy/scenario switching without changing the core model structure.

5.3 3) Primary Reserve (if active)

Description: The primary reserve module enforces hourly up/down reserve requirements and limits reserve provision using unit headroom/footroom and storage/flexibility feasibility conditions. **Role:** enforces short-term balancing capability.

5.4 4) Secondary Reserve (if active)

Description: This block mirrors the primary reserve logic with dedicated variables and constraints for secondary reserve products. **Role:** enforces additional balancing margin.

5.5 5) Industrial Heat Demand and Fuel Switching

Description: The equations below set a fuel-share upper bound per fuel and period, then close the industrial heat balance with endogenous hydrogen substitution.

$$\texttt{Fuel\_ind}_{f,t,y} \leq \texttt{fuel\_share\_max}_{t,y,f} \cdot \left(\texttt{Industry\_HT\_demand}_{t,y} - \texttt{H2tInd}_{t,y}\right) \tag{10}$$

$$\sum_{f} \texttt{Fuel\_ind}_{f,t,y} + \texttt{H2tInd}_{t,y} = \texttt{Industry\_HT\_demand}_{t,y} \tag{11}$$

Role: satisfies industrial thermal demand while allowing endogenous fuel mix with hydrogen substitution.

5.6 6) Hydrogen Production, Storage, and Demand Balance

Description: This block links electricity use to hydrogen production via efficiency, limits hydrogen conversion capacity, updates hydrogen storage SOC, and balances total hydrogen withdrawals with final uses.

$$\sum_{e \in \texttt{Elec\_h2\_units}} \frac{\texttt{generation\_H2}_{e,t,y}}{\texttt{H2\_eff}_{e}} = \texttt{power\_to\_h2}_{t,y} \tag{12}$$

$$\texttt{generation\_H2}_{e,t,y} \leq \texttt{H2\_init\_cap}_{e} + \sum_{\tau \leq y} \texttt{h2\_electrolysis\_max\_cap}_{e,\tau} \tag{13}$$

$$\texttt{h2\_sto\_soc}_{s,t,y} = \texttt{h2\_sto\_soc}_{s,t-1,y} - \texttt{h2\_sto\_out}_{s,t,y} + \sum_{e} \texttt{generation\_H2}_{e,t,y} \tag{14}$$

with SOC bounds and total demand balance:

$$\sum_{s} \texttt{h2\_sto\_out}_{s,t,y} = \sum_{q} \texttt{H2\_demand}_{q,t,y} + \sum_{c} \frac{\texttt{generation\_FC}_{c,t,y}}{\texttt{FC\_eff}_{c}} + \texttt{H2tInd}_{t,y} \tag{15}$$

Role: ensures physical consistency of the hydrogen subsystem.

5.7 7) EV and Stationary Storage Dynamics

Description: These equations track EV and stationary storage SOC trajectories over time with efficiency-adjusted charge/discharge flows, while enforcing SOC bounds, power limits, investment-based capacity limits, and end-of-horizon conditions.

5.7.1 7.1 EV Fleet Storage

Compact notation:

$$E^{ev}_{t,y} \equiv \texttt{ev\_sto\_soc}_{t,y}, \quad P^{ev,in}_{t,y} \equiv \texttt{ev\_sto\_in}_{t,y}, \quad P^{ev,out}_{t,y} \equiv \texttt{ev\_sto\_out}_{t,y},$$
$$\eta^{ev} \equiv \texttt{ev\_Grid\_eff}, \quad \overline{E}^{ev}_{y} \equiv \texttt{ev\_sto\_max}_{y}, \quad \alpha^{ev} \equiv \texttt{ev\_sto\_min},$$
$$\overline{P}^{ev}_{y} \equiv \texttt{ev\_Grid\_P\_max}_{y}, \quad a^{ev}_{t,y} \equiv \texttt{ev\_availability}_{t,y}, \quad L^{ev}_{t,y} \cdot D^{ev}_{y} \equiv \texttt{ev\_transp\_load}_{t,y} \cdot \texttt{ev\_Demand}_{y}.$$

$$E^{ev}_{1,y} = -L^{ev}_{1,y} \cdot D^{ev}_{y} - P^{ev,out}_{1,y} + \eta^{ev} \cdot P^{ev,in}_{1,y} \tag{16}$$

$$E^{ev}_{t,y} = E^{ev}_{t-1,y} - L^{ev}_{t,y} \cdot D^{ev}_{y} - P^{ev,out}_{t,y} + \eta^{ev} \cdot P^{ev,in}_{t,y} \tag{17}$$

$$\alpha^{ev} \cdot \overline{E}^{ev}_{y} \le E^{ev}_{t,y} \le \overline{E}^{ev}_{y} \tag{18}$$

$$P^{ev,out}_{t,y} \le \overline{P}^{ev}_{y} \cdot a^{ev}_{t,y} \tag{19}$$
$$P^{ev,in}_{t,y} \le \overline{P}^{ev}_{y} \cdot a^{ev}_{t,y} \tag{20}$$

5.7.2 7.2 Stationary Battery Storage

Compact notation:

$$E^{st}_{b,t,y} \equiv \texttt{stat\_sto\_soc}_{b,t,y}, \quad P^{st,in}_{b,t,y} \equiv \texttt{stat\_sto\_in}_{b,t,y},$$
$$P^{st,out}_{b,t,y} \equiv \texttt{stat\_sto\_out}_{b,t,y}, \quad \eta^{st}_{b} \equiv \texttt{Sta\_sto\_eff}_{b},$$
$$\overline{E}^{st}_{b,y} \equiv \texttt{Sta\_sto\_init\_cap}_{b} \cdot \texttt{StaSto\_Decom\_ind}_{y} + \sum_{\tau \le y} \texttt{stat\_sto\_cap\_inv}_{b,\tau} \cdot \texttt{StaStoDecom}_{b,\tau,y}.$$

$$E^{st}_{b,1,y} = -P^{st,out}_{b,1,y} + \eta^{st}_{b} \cdot P^{st,in}_{b,1,y} \tag{21}$$

$$E^{st}_{b,t,y} = E^{st}_{b,t-1,y} - P^{st,out}_{b,t,y} + \eta^{st}_{b} \cdot P^{st,in}_{b,t,y} \tag{22}$$

$$E^{st}_{b,t,y} \le \overline{E}^{st}_{b,y} \tag{23}$$

$$\sum_{y} \texttt{stat\_sto\_cap\_inv}_{b,y} \le \texttt{Sta\_sto\_max\_cap}_{b} \tag{24}$$

$$E^{st}_{b,T,y} \le 0.2\,\texttt{Sta\_sto\_max\_cap}_{b} \tag{25}$$

Role: shifts load and supports balancing through electrochemical storage.

5.8 8) Generation Limits by Technology Type

Description: The block first defines effective available capacity from surviving existing assets plus surviving new investments, then applies technology-specific dispatch rules and annual investment ceilings.

$$\texttt{CapEff}_{u,y} = \texttt{max\_load}_u \cdot \texttt{DecomOld}_{u,y} + \sum_{\tau \le y} \texttt{cap\_inv}_{u,\tau} \cdot \texttt{DecomNew}_{u,\tau,y} \tag{26}$$

$$\texttt{generation}_{u,t,y} \le \texttt{cap\_factor}_u \cdot \texttt{CapEff}_{u,y}, \quad u \in \texttt{fossil\_units} \cup \texttt{hphs\_units} \cup \texttt{hdam\_units} \tag{27}$$

$$\texttt{generation}_{u,t,y} = \texttt{aval\_factor\_plant}_{u,t,y} \cdot \texttt{CapEff}_{u,y}, \quad u \in \texttt{nondisp\_units} \cup \texttt{hror\_units} \tag{28}$$

$$\texttt{cap\_inv}_{u,y} \le \texttt{max\_inv\_period}_u \tag{29}$$

Role: enforces physical and planning feasibility of generation.

5.9 9) Electricity Balance and CEPP Bound

Description: This block enforces hourly electricity adequacy by balancing injections (generation, imports, and storage discharge) with withdrawals (demand, charging, and sector-coupling electric loads). It also bounds annual `ceep` as a fraction of yearly electric-energy requirements. *Compact notation:*

$$\begin{aligned}
\mathcal{U}^{el} &\equiv \texttt{fossil_units} \cup \texttt{hphs_units} \cup \texttt{hdam_units} \cup \texttt{hror_units} \cup \texttt{nondisp_units}, \\
G^{tot}_{t,y} &\equiv \texttt{GenTot}_{t,y}, \quad G_{u,t,y} \equiv \texttt{generation}_{u,t,y}, \quad G^{FC}_{c,t,y} \equiv \texttt{generation_FC}_{c,t,y}, \\
I_{t,y} &\equiv \texttt{imports}_{t,y}, \quad D_{s,t,y} \equiv \texttt{demand}_{s,t,y}, \quad C_{t,y} \equiv \texttt{ceep}_{t,y}, \\
P^{ev,in}_{t,y} &\equiv \texttt{ev_sto_in}_{t,y}, \quad P^{ev,out}_{t,y} \equiv \texttt{ev_sto_out}_{t,y}, \quad \eta^{ev} \equiv \texttt{ev_Grid_eff}, \\
P^{st,in}_{b,t,y} &\equiv \texttt{stat_sto_in}_{b,t,y}, \quad P^{st,out}_{b,t,y} \equiv \texttt{stat_sto_out}_{b,t,y}, \quad \eta^{st}_b \equiv \texttt{Sta_sto_eff}_b, \\
H_{t,y} &\equiv \texttt{power_to_h2}_{t,y}, \quad R_{t,y} \equiv \texttt{ResToHeat}_{t,y}, \quad \bar{\alpha}^{ceep} \equiv \texttt{ceep_parameter}, \\
N^{el}_{t,y} &\equiv \texttt{noload_elec}_{t,y}.
\end{aligned}$$

$$G^{tot}_{t,y} = \sum_{u \in \mathcal{U}^{el}} G_{u,t,y} + \sum_{c \in \texttt{FC\_units}} G^{FC}_{c,t,y} \tag{30}$$

$$G^{tot}_{t,y} + I_{t,y} + \eta^{ev} \cdot P^{ev,out}_{t,y} + \sum_b \eta^{st}_b \cdot P^{st,out}_{b,t,y} + N^{el}_{t,y} = \sum_s D_{s,t,y} + \sum_b P^{st,in}_{b,t,y} + P^{ev,in}_{t,y} + H_{t,y} + R_{t,y} + C_{t,y} \tag{31}$$

$$\sum_t C_{t,y} \le \bar{\alpha}^{ceep} \cdot \Big(\sum_{s,t} D_{s,t,y} + \sum_{b,t} P^{st,in}_{b,t,y} + \sum_t R_{t,y} + \sum_t H_{t,y} + \sum_t P^{ev,in}_{t,y} \Big) \tag{32}$$

Role: central electricity adequacy condition of the integrated system, plus an explicit annual quality/reliability bound through limited `ceep`. In the implemented v1.1 code, adequacy can also be relaxed through the explicit variable `noload_elec`, and the annual CEEP denominator is evaluated against net storage charging rather than gross charging alone.

5.10 10) Heat/Cooling Block (HP, Boilers, CHP)

Description: This block maps electricity-to-heat conversion via COP relations, enforces heat-pump and boiler operating limits, captures CHP heat-power coupling, updates thermal storage states, and closes hourly heat/cooling demand balances. *Compact notation:*

$$
\begin{aligned}
Q^{dh}_{h,m,t,y} &\equiv \texttt{heat\_pump\_dh}_{h,m,t,y}, & Q^{cdh}_{h,m,t,y} &\equiv \texttt{cooling\_pump\_dh}_{h,m,t,y},\\
Q^{ind}_{h,t,y} &\equiv \texttt{heat\_pump\_ind}_{h,t,y}, & Q^{cind}_{h,t,y} &\equiv \texttt{cooling\_pump\_ind}_{h,t,y},\\
Q^{out}_{h,t,y} &\equiv \texttt{heat\_pump\_out}_{h,t,y}, & E^{hp}_{h,t,y} &\equiv \texttt{heat\_pump\_sto\_soc}_{h,t,y},\\
COP_{h,t,y} &\equiv \texttt{HP\_COP\_period}_{h,t,y}, & R_{t,y} &\equiv \texttt{ResToHeat}_{t,y},\\
\bar{Q}^{dh}_{h,m,y} &\equiv \texttt{HPDHCapEff}_{h,m,y}, & \bar{Q}^{ind}_{h,y} &\equiv \texttt{HPINDCapEff}_{h,y},\\
B^{dh}_{b,m,t,y} &\equiv \texttt{boiler\_dh\_generation}_{b,m,t,y}, & B^{ind}_{b,t,y} &\equiv \texttt{boiler\_ind\_generation}_{b,t,y},\\
\bar{B}^{dh}_{b,m,y} &\equiv \texttt{BoilerDHCapEff}_{b,m,y}, & \bar{B}^{ind}_{b,y} &\equiv \texttt{BoilerIndCapEff}_{b,y},\\
H^{in}_{m,t,y} &\equiv \texttt{heat\_input}_{m,t,y}, & G_{m,t,y} &\equiv \texttt{generation}_{m,t,y},\\
\kappa^{ht}_{m} &\equiv \texttt{CHPPowerToHeat}_{m}, & \kappa^{loss}_{m} &\equiv \texttt{CHPPowerLossFactor}_{m},\\
\bar{G}_{m} &\equiv \texttt{max\_load}_{m} \cdot \texttt{cap\_factor}_{m}, & S_{m,t,y} &\equiv \texttt{heat\_sto\_level}_{m,t,y},\\
Q^{gen}_{m,t,y} &\equiv \texttt{heat\_gen}_{m,t,y}, & D^{heat}_{m,t,y} &\equiv \texttt{heat\_demand}_{m,t,y},\\
D^{heat,g}_{t,y} &\equiv \texttt{heat\_demand}_{g^{heat},t,y}, & D^{cool,g}_{t,y} &\equiv \texttt{cooling\_demand}_{g^{cool},t,y},\\
N^{dh}_{m,t,y} &\equiv \texttt{noload\_heat\_dh}_{m,t,y}, & N^{ind}_{t,y} &\equiv \texttt{noload\_heat\_ind}_{t,y},\\
g^{heat} &\equiv \texttt{general\_demand}, & g^{cool} &\equiv \texttt{general\_cooling}.
\end{aligned}
$$

$$\sum_{h,m} \frac{Q^{dh}_{h,m,t,y} + Q^{cdh}_{h,m,t,y}}{COP_{h,t,y}} + \sum_{h} \frac{Q^{ind}_{h,t,y}}{COP_{h,t,y}} = R_{t,y} \tag{33}$$

$$Q^{dh}_{h,m,t,y} + Q^{cdh}_{h,m,t,y} \leq \bar{Q}^{dh}_{h,m,y} \tag{34}$$

$$Q^{ind}_{h,t,y} \leq \bar{Q}^{ind}_{h,y} \tag{35}$$

$$E^{hp}_{h,t,y} = E^{hp}_{h,t-1,y} + Q^{ind}_{h,t,y} - Q^{out}_{h,t,y} - Q^{cind}_{h,t,y} \tag{36}$$

$$B^{dh}_{b,m,t,y} \leq \bar{B}^{dh}_{b,m,y} \tag{37}$$

$$B^{ind}_{b,t,y} \leq \bar{B}^{ind}_{b,y} \tag{38}$$

$$H^{in}_{m,t,y} \cdot \kappa^{ht}_{m} \leq G_{m,t,y} \tag{39}$$

$$G_{m,t,y} \leq \bar{G}_{m} - H^{in}_{m,t,y} \cdot \kappa^{loss}_{m} \tag{40}$$

$$S_{m,t,y} = S_{m,t-1,y} + H^{in}_{m,t,y} + \sum_{h} Q^{dh}_{h,m,t,y} + \sum_{b} B^{dh}_{b,m,t,y} - Q^{gen}_{m,t,y} \tag{41}$$

$$Q^{gen}_{m,t,y} + N^{dh}_{m,t,y} = D^{heat}_{m,t,y} \tag{42}$$

$$\sum_h Q^{out}_{h,t,y} + \sum_b B^{ind}_{b,t,y} + N^{ind}_{t,y} = D^{heat,g}_{t,y} \tag{43}$$

$$\sum_h Q^{cind}_{h,t,y} = D^{cool,g}_{t,y} \tag{44}$$

Role: ensures thermally feasible operation and sector coupling consistency while preserving hourly heat/cooling service adequacy. The current v1.1 code additionally blocks cooling provision from electric-boiler variables and applies the CHP electric upper bound using surviving effective capacity rather than a purely static nameplate envelope.

5.11 11) Hydro Reservoir and Pumped Hydro Storage

Description: This block models reservoir and pumped-hydro storage over time. It enforces storage bounds, water-balance recursions with inflow and efficiency, feasible withdrawals (generation plus spill), and end-of-horizon storage targets. *Compact notation:*

$\mathcal{U}^{dam} \equiv \texttt{hdam\_units}$, $\mathcal{U}^{phs} \equiv \texttt{hphs\_units}$, $G_{u,t,y} \equiv \texttt{generation}_{u,t,y}$,
$S_{u,t,y} \equiv \texttt{storage\_level}_{u,t,y}$, $P_u \equiv \texttt{max\_load}_u$, $O_{u,t,y} \equiv \texttt{sto\_out\_flow}_{u,t,y}$,
$F^{dam}_{u,t,y} \equiv \texttt{hdam\_inflow}_{u,t,y}$, $\eta^{dam}_u \equiv \texttt{hdam\_efficiency}_u$, $\bar{S}^{dam}_u \equiv \texttt{hdam\_max\_storage}_u$,
$F^{phs}_{u,t,y} \equiv \texttt{hphs\_inflow}_{u,t,y}$, $\eta^{phs}_u \equiv \texttt{hphs\_efficiency}_u$, $\bar{S}^{phs}_u \equiv \texttt{hphs\_max\_storage}_u$.

$$0.5\,\bar{S}^{dam}_u \le S_{u,t,y} \le \bar{S}^{dam}_u, \quad u \in \mathcal{U}^{dam} \tag{45}$$

$$S_{u,1,y} = 0.5\,\bar{S}^{dam}_u + F^{dam}_{u,1,y} \cdot P_u \cdot \eta^{dam}_u - \frac{G_{u,1,y}}{\eta^{dam}_u} - O_{u,1,y} \tag{46}$$

$$S_{u,t,y} = S_{u,t-1,y} + F^{dam}_{u,t,y} \cdot P_u \cdot \eta^{dam}_u - \frac{G_{u,t,y}}{\eta^{dam}_u} - O_{u,t,y} \tag{47}$$

$$\frac{G_{u,t,y}}{\eta^{dam}_u} + O_{u,t,y} \le S_{u,t,y} + F^{dam}_{u,t,y} \cdot P_u \cdot \eta^{dam}_u \tag{48}$$

$$S_{u,T,y} \ge 0.5\,\bar{S}^{dam}_u \tag{49}$$

$$0.5\,\bar{S}^{phs}_u \le S_{u,t,y} \le \bar{S}^{phs}_u, \quad u \in \mathcal{U}^{phs} \tag{50}$$

$$S_{u,t,y} = S_{u,t-1,y} + F^{phs}_{u,t,y} \cdot P_u \cdot \eta^{phs}_u - \frac{G_{u,t,y}}{\eta^{phs}_u} - O_{u,t,y} \tag{51}$$

$$\frac{G_{u,t,y}}{\eta^{phs}_u} + O_{u,t,y} \le S_{u,t,y} + F^{phs}_{u,t,y} \cdot P_u \cdot \eta^{phs}_u \tag{52}$$

$$S_{u,T,y} \ge 0.5\,\bar{S}^{phs}_u \tag{53}$$

Role: preserves physically feasible hydro energy trajectories and prevents end-horizon reservoir depletion.

5.12 12) Ramping, Imports, RPS, and CO2 Policy

Description: This block combines short-term operational constraints (ramping and import smoothing) with strategic policy constraints (renewable portfolio share and annual CO2 cap). *Compact notation:*

$$\begin{aligned}
\mathcal{U}^{disp} &\equiv \texttt{disp_units}, \quad \mathcal{U}^{rps} \equiv \texttt{nondisp_units} \cup \texttt{biomass_units} \cup \texttt{hydro_units}, \\
\mathcal{U}^{el} &\equiv \texttt{fossil_units} \cup \texttt{hphs_units} \cup \texttt{hdam_units} \cup \texttt{hror_units} \cup \texttt{nondisp_units}, \\
G_{u,t,y} &\equiv \texttt{generation}_{u,t,y}, \quad I_{t,y} \equiv \texttt{imports}_{t,y}, \quad \bar{I}_t \equiv \texttt{import_ntc}_t, \\
r_u^{up} &\equiv \texttt{ramp_up_rate}_u, \quad r_u^{down} \equiv \texttt{ramp_down_rate}_u, \quad \rho_y \equiv \texttt{rps}_y, \\
E_y^{el} &\equiv \texttt{CO2_total}_y, \quad E_y^{ht} \equiv \texttt{CO2_total_heat}_y, \quad E_y^{ind} \equiv \texttt{CO2_total_industry}_y, \\
\bar{E}_y &\equiv \texttt{CO2_limit}_y, \quad \phi_u \equiv \texttt{CO2_factor}_u, \quad \phi_b^{th} \equiv \texttt{Thermal_CO2}_b, \\
\phi_f^f &\equiv \texttt{Fuel_CO2}_f, \quad B_{b,t,y}^{ind} \equiv \texttt{boiler_ind_generation}_{b,t,y}, \\
B_{b,m,t,y}^{dh} &\equiv \texttt{boiler_dh_generation}_{b,m,t,y}, \quad F_{f,t,y} \equiv \texttt{Fuel_ind}_{f,t,y}.
\end{aligned}$$

$$G_{u,t,y} \le G_{u,t-1,y} + r_u^{up} \cdot \texttt{CapEff}_{u,y}, \quad u \in \mathcal{U}^{disp} \tag{54}$$

$$G_{u,t,y} \ge G_{u,t-1,y} - r_u^{down} \cdot \texttt{CapEff}_{u,y}, \quad u \in \mathcal{U}^{disp} \tag{55}$$

$$I_{t,y} \le \bar{I}_t \tag{56}$$

$$I_{t,y} \le 1.2\, I_{t-1,y} \tag{57}$$

$$I_{t,y} \ge 0.8\, I_{t-1,y} \tag{58}$$

$$\sum_{u \in \mathcal{U}^{rps}} \sum_t G_{u,t,y} \ge \rho_y \left(\sum_{u \in \mathcal{U}^{el}} \sum_t G_{u,t,y} + \sum_t I_{t,y} \right) \tag{59}$$

$$E_y^{el} = \sum_{u \in \texttt{fossil\_units}, t} \phi_u \cdot G_{u,t,y} \tag{60}$$

$$E_y^{ht} = \sum_{b,t} \phi_b^{th} \cdot B_{b,t,y}^{ind} + \sum_{b,m,t} \phi_b^{th} \cdot B_{b,m,t,y}^{dh} \tag{61}$$

$$E_y^{ind} = \sum_{f,t} \phi_f^f \cdot F_{f,t,y} \tag{62}$$

$$E_y^{el} + E_y^{ht} + E_y^{ind} \le \bar{E}_y \tag{63}$$

Role: preserves feasible system operation hour-by-hour while simultaneously enforcing portfolio and emissions policy trajectories.

5.13 13) Ramping Cost Linearization

Description: Auxiliary ramp-cost variables are lower-bounded by signed generation changes, yielding a linear representation of up/down ramping penalties in the objective.

$$\texttt{rampup\_cost}_{u,t,y} \geq \texttt{ramp\_cost}_u \left(G_{u,t,y} - G_{u,t-1,y}\right) \tag{64}$$

$$\texttt{rampdown\_cost}_{u,t,y} \geq \texttt{ramp\_cost}_u \left(-G_{u,t,y} + G_{u,t-1,y}\right) \tag{65}$$

Role: penalizes fast dispatch changes while preserving linearity.

6 Implementation Notes (Current Code Traceability)

- `Power_to_h2` is created with an extra loop over `flex_unit`; mathematically it is equivalent to one equation per (t, y) but instantiated repeatedly.
- `primary_hourly_reserve_down_units_minimum2` and `secondary_hourly_reserve_down_units_minimum2` use an up-reserve variable on the left-hand side.
- `CO2_cost_industry` is computed but not included in the final objective sum.
- With `Imp_price_inc=0.05`, the expression $(1 + \texttt{Imp\_price\_inc}/100)^n$ corresponds to 0.05% annual growth.
- The v1.1 code introduces explicit unmet-service variables `noload_elec`, `noload_heat_ind`, and `noload_heat_dh`, together with the penalty terms `ceep_cost`, `no_load_cost`, and `no_load_cost_dh`.
- Input preprocessing in v1.1 migrates from SpreadsheetML/XML parsing to direct CSV/XLSX ingestion, while preserving the same identifier-matching logic across `year`, `period`, and unit or fuel labels.
- The current v1.1 script defines separate annual CAPEX dictionaries for electrolyzers, fuel cells, heat pumps, boilers, hydrogen storage, and stationary batteries instead of relying on a single generic flex-technology cost table.
- The present v1.1 code also sets `periods = list(range(1,10))`; treat this as an implementation-specific reduced horizon unless it is intentionally retained for the study design.

A Appendix A: Reserve Module Data Dictionary

This appendix consolidates the data dictionary entries used specifically in Sections 4.3 and 4.4.

A.1 A.1 Decision Variables for Reserve Modules

Variable (original model notation)	Type	Unit	Description
$\texttt{primary\_hourly\_reserve\_up\_units}_{u,t,y}$	Reserve	MW_e	Upward primary reserve from dispatchable units.
$\texttt{primary\_hourly\_reserve\_down\_units}_{u,t,y}$	Reserve	MW_e	Downward primary reserve from dispatchable units.
$\texttt{primary\_hourly\_stat\_stoOUT\_reserve\_up\_units}_{b,t,y}$	Reserve	MW_e	Upward primary reserve via stationary storage discharge margin.
$\texttt{primary\_hourly\_stat\_stoOUT\_reserve\_down\_units}_{b,t,y}$	Reserve	MW_e	Downward primary reserve via stationary storage discharge margin.
$\texttt{primary\_hourly\_stat\_stoIN\_reserve\_up\_units}_{b,t,y}$	Reserve	MW_e	Upward primary reserve via stationary storage charging margin.
$\texttt{primary\_hourly\_stat\_stoIN\_reserve\_down\_units}_{b,t,y}$	Reserve	MW_e	Downward primary reserve via stationary storage charging margin.
$\texttt{primary\_hourly\_FC\_reserve\_up}_{c,t,y}$	Reserve	MW_e	Upward primary reserve from fuel cells.
$\texttt{primary\_hourly\_FC\_reserve\_down}_{c,t,y}$	Reserve	MW_e	Downward primary reserve from fuel cells.
$\texttt{primary\_hourly\_ELY\_reserve\_down}_{e,t,y}$	Reserve	MW_e	Downward primary reserve from electrolyzers.
$\texttt{primary\_hourly\_ELY\_reserve\_up}_{e,t,y}$	Reserve	MW_e	Upward primary reserve from electrolyzers.
$\texttt{primary\_hourly\_HPind\_reserve\_down}_{h,t,y}$	Reserve	MW_e	Downward primary reserve from industrial heat pumps.
$\texttt{primary\_hourly\_HPind\_reserve\_up}_{h,t,y}$	Reserve	MW_e	Upward primary reserve from industrial heat pumps.
$\texttt{secondary\_hourly\_reserve\_up\_units}_{u,t,y}$	Reserve	MW_e	Upward secondary reserve from dispatchable units.
$\texttt{secondary\_hourly\_reserve\_down\_units}_{u,t,y}$	Reserve	MW_e	Downward secondary reserve from dispatchable units.
$\texttt{secondary\_hourly\_stat\_stoOUT\_reserve\_up\_units}_{b,t,y}$	Reserve	MW_e	Upward secondary reserve via stationary storage discharge margin.
$\texttt{secondary\_hourly\_stat\_stoOUT\_reserve\_down\_units}_{b,t,y}$	Reserve	MW_e	Downward secondary reserve via stationary storage discharge margin.
$\texttt{secondary\_hourly\_stat\_stoIN\_reserve\_up\_units}_{b,t,y}$	Reserve	MW_e	Upward secondary reserve via stationary storage charging margin.
$\texttt{secondary\_hourly\_stat\_stoIN\_reserve\_down\_units}_{b,t,y}$	Reserve	MW_e	Downward secondary reserve via stationary storage charging margin.
$\texttt{secondary\_hourly\_FC\_reserve\_up}_{c,t,y}$	Reserve	MW_e	Upward secondary reserve from fuel cells.
$\texttt{secondary\_hourly\_FC\_reserve\_down}_{c,t,y}$	Reserve	MW_e	Downward secondary reserve from fuel cells.
$\texttt{secondary\_hourly\_ELY\_reserve\_down}_{e,t,y}$	Reserve	MW_e	Downward secondary reserve from electrolyzers.
$\texttt{secondary\_hourly\_ELY\_reserve\_up}_{e,t,y}$	Reserve	MW_e	Upward secondary reserve from electrolyzers.
$\texttt{secondary\_hourly\_HPind\_reserve\_down}_{h,t,y}$	Reserve	MW_e	Downward secondary reserve from industrial heat pumps.
$\texttt{secondary\_hourly\_HPind\_reserve\_up}_{h,t,y}$	Reserve	MW_e	Upward secondary reserve from industrial heat pumps.

A.2 A.2 Parameters and Sets for Reserve Modules

Parameter / set (original model notation)	Category	Unit	Description
`PrimaryReserve`	Scenario flag	bool	Activates the primary reserve constraint block.
`SecondaryReserve`	Scenario flag	bool	Activates the secondary reserve constraint block.
`res_inv`	Scenario flag	bool	Activates reserve-investment related module switches.
`reserve_factor`$_{u,t,y}$	Technical	p.u.	Reserve-down scaling factor applied at the same hourly granularity as reserve variables (often time-invariant in data).
`primary_reserve_units`	Set	-	Index set for u in primary reserve variables `primary_hourly_reserve_up_units`$_{u,t,y}$ and `primary_hourly_reserve_down_units`$_{u,t,y}$.
`primary_reserve_StatStorage_units`	Set	-	Index set for b in stationary-storage primary reserve variables.
`primary_reserve_FC_units`	Set	-	Index set for c in fuel-cell primary reserve variables.
`primary_reserve_ELY_units`	Set	-	Index set for e in electrolyzer primary reserve variables.
`primary_reserve_HP_units`	Set	-	Index set for h in industrial heat-pump primary reserve variables.
`secondary_reserve_units`	Set	-	Index set for u in secondary reserve variables `secondary_hourly_reserve_up_units`$_{u,t,y}$ and `secondary_hourly_reserve_down_units`$_{u,t,y}$.
`secondary_reserve_StatStorage_units`	Set	-	Index set for b in stationary-storage secondary reserve variables.
`secondary_reserve_FC_units`	Set	-	Index set for c in fuel-cell secondary reserve variables.
`secondary_reserve_ELY_units`	Set	-	Index set for e in electrolyzer secondary reserve variables.
`secondary_reserve_HP_units`	Set	-	Index set for h in industrial heat-pump secondary reserve variables.